SESSION 4

Classification Fundamentals

Logistic regression • Decision trees • Confusion matrices • Precision • Recall • F1-score

Applied AI Design Workshop for Faculty

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Goal: build intuition for predicting categories and choosing metrics based on the cost of errors.

Session focus
- Classification problem framing
- Probabilities and thresholds
- Logistic regression and trees
- Confusion matrices
- Precision, recall, and F1

Session 4

From engineering problem to ML problem

Classic ML overview

Invited presentation: From Prompts to Agents

Break

Regression fundamentals

Classification fundamentals

Hands-on baseline model activity

Earlier
Problem framing
Regression

Now

Next
Hands-on
baseline model

Session 4 learning outcomes

Frame
Recognize when a categorical target makes classification appropriate

Explain
Describe logistic regression, decision trees, and predicted probabilities

Evaluate
Interpret confusion matrices and common classification metrics

Choose
Select metrics based on consequences of false positives and false negatives

Prepare
Connect classification concepts to the baseline model activity

Participant output: a classification problem statement, a baseline choice, an initial metric, and one discussion of error consequences.

What makes a problem a classification problem?

Classification predicts a category or class label. The model is judged by whether the predicted class is correct and which errors matter most.

Use classification when asking:

- Which category?
- Pass or fail?
- Normal or faulty?
- Eligible or not?
- Which condition?

Typical targets:

- Defective / acceptable
- Fault / no fault
- Risk level
- Material type
- Operating mode
- Course outcome

Not classification when:

- Target is continuous
- Goal is discovering unknown groups
- No labeled examples exist
- The output is a ranking only

Key discipline question: Which error is more costly — a false alarm or a missed case?

Activity: write a classification problem statement

Use the template to convert a disciplinary problem into a classification task.

Using __________ as inputs, develop a classification model to predict __________ for each __________, with performance evaluated using __________.

Inputs Variables available before prediction	**Class label** Category or state to predict	**Unit** Part, sample, student, machine cycle, image, day	**Metric** Accuracy, precision, recall, F1, or class-specific target

Quick check: are the labels trustworthy, consistent, and available for enough examples?

Example: product quality classification

A classification dataset pairs measured features with a categorical label such as acceptable or defective.

Part	Temp	Speed	Vibration	Roughness	Label
1	185	42	0.18	2.4	Accept
2	190	46	0.35	4.2	Defect
3	174	39	0.22	2.9	Accept
4	205	52	0.48	5.1	Defect
5	198	44	0.20	2.5	Accept

Prediction objective
Classify whether a part is acceptable before final inspection.

Baseline
Predict the majority class for every part.

Metric
Use recall for defects when missed defects are expensive.

Start with a classification baseline

A baseline clarifies whether the classifier adds value beyond the simplest possible decision rule.

Majority-class baseline

Predict the most frequent class for every case.

Essential when classes are imbalanced.

Random or stratified baseline

Predict classes based on observed class proportions.

Useful for sanity checking.

Domain-rule baseline

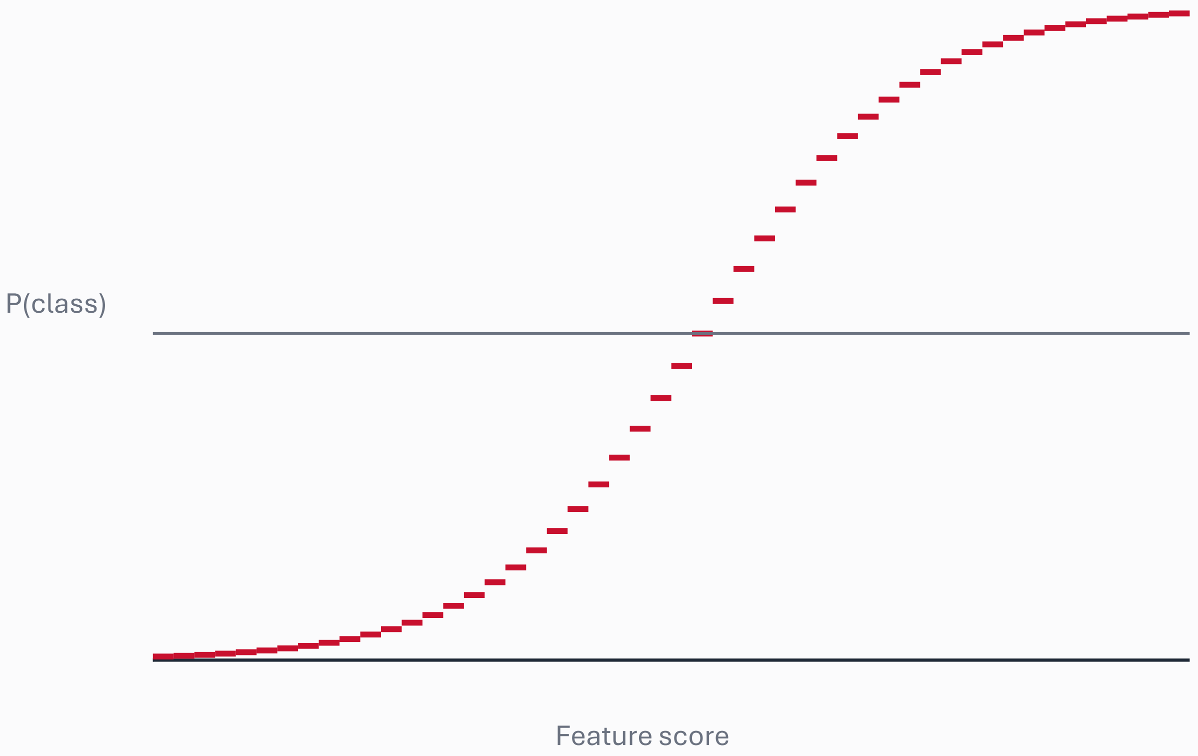
Use a threshold, checklist, expert rule, or existing procedure.

Often the fairest comparison.

Discussion prompt: Would stakeholders accept a model that beats accuracy but misses the most important cases?

Logistic regression: probabilities for class membership

Logistic regression estimates the probability that an observation belongs to a class, then applies a decision threshold.



Decision rule
If P(defect) ≥ threshold → classify as defect

Interpretation
Coefficients describe how features shift log-odds, not direct class labels.

Caution
The default 0.50 threshold may not match the cost of errors.

Decision trees: interpretable rule-based classification

Decision trees split the data into increasingly pure groups using feature-based rules.

Root question	Branch question	Leaf	Leaf
Is vibration > 0.30?	Is roughness > 3.5?	Defect	Accept

A path through the tree becomes a rule: if vibration is high, flag defect; otherwise inspect roughness.

Strength
Easy to explain as a sequence of rules.

Risk
Deep trees can overfit training data.

Control
Limit depth, require minimum samples per leaf, or prune.

Confusion matrix: the basic language of classification error

A confusion matrix separates correct predictions from different types of mistakes.

		Predicted class	
		Defect	Accept
Actual class	Defe ct	TP Detected defects	FN Missed defects
	Acce pt	FP False alarms	TN Correct accepts

False negative: a defect is missed.

False positive: an acceptable part is flagged.

Different applications assign different costs to these errors.

Classification metrics: accuracy is not enough

Metrics should match the consequences of different mistakes, especially for imbalanced classes.

Accuracy	**Overall percent correct**	Can be misleading when one class dominates
Precision	**Of predicted positives, how many were correct?**	Use when false alarms are costly
Recall	**Of actual positives, how many were found?**	Use when missed cases are costly
F1-score	**Balance of precision and recall**	Useful when both false alarms and misses matter

Metric choice is a design decision: it defines which mistakes the model is allowed to make.

Class imbalance: when accuracy hides failure

A model can appear highly accurate by ignoring rare but important classes.

Scenario

1000 parts
950 acceptable
50 defective

Majority baseline predicts every part as acceptable.

Result

Accuracy = 95%
Defect recall = 0%

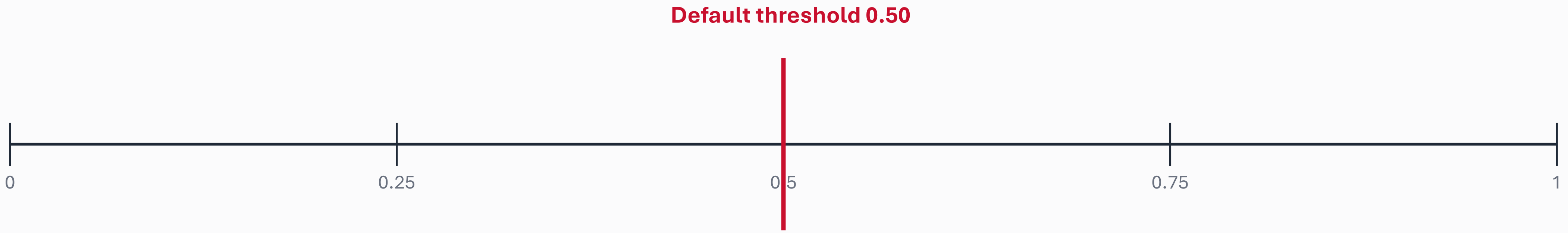
The model misses every defective part.

Better questions

- How many defects are found?
- How many false alarms occur?
- What threshold should trigger review?

Decision thresholds: choosing where to draw the line

Many classifiers return probabilities. Changing the threshold changes false positives and false negatives.



Lower threshold
More positives flagged
Higher recall
More false alarms

Higher threshold
Fewer positives flagged
Higher precision
More missed cases

Best threshold
Depends on cost, capacity, risk, and review process

Activity: interpret model outputs

Compare two classifiers and decide which is more appropriate for quality inspection.

Model A

Accuracy: 94%
Precision: 70%
Recall: 40%
F1-score: 51%

Model B

Accuracy: 91%
Precision: 55%
Recall: 82%
F1-score: 66%

Decision

Which model should be selected if missed defects are costly?

Which if false alarms are costly?

Debrief: a lower accuracy model may be the better disciplinary choice when it reduces the more serious error.

Classification Examples Datasets

Manufacturing	Classify product acceptability, defect type, machine state, or process condition from measurements.
Bioprocessing	Classify growth phase, contamination risk, or abnormal operating state from sensor summaries.
Buildings	Classify occupancy state, energy anomaly, or equipment fault from time and sensor data.
Education	Classify early support need or course outcome from activity and performance indicators.
Equipment health	Classify normal, warning, and fault states from vibration, current, temperature, and runtime.

Classroom discussion prompts and transition

Why can accuracy be misleading?

Which error is more costly: false positive or false negative?

Should the threshold always be 0.50?

When is a decision tree preferable to logistic regression?

How trustworthy are the class labels?

What baseline represents current practice?

Session synthesis: classification is not just about predicting labels — it is about choosing errors that align with disciplinary consequences.

Next: Hands-on baseline model activity — apply regression or classification to a small dataset.